1.

```
def largeGroupPositions(s):
    result = []
    i = 0

    while i < len(s):
        j = i

        while j < len(s) and s[j] == s[i]:
            j += 1

        if j - i >= 3:
            result.append([i, j - 1])

        i = j

    return result

s = input().strip()
print(largeGroupPositions(s))
```

EXAMPLE 1:

```
= RESTART: C:/Users/miche/OneDrive/Desktop/DAA CLG LAB/11-10-26/Large Groups in
a String.py
abbxxxxzzy
[[3, 6]]
>>>
```

EXAMPLE 2:

```
= RESTART: C:/Users/miche/OneDrive/Desktop/DAA CLG LAB/11-10-26/Large Groups in
a String.py
a b c
[]
>>
```

2.

```
def gameOfLife(board):
    m = len(board)
    n = len(board[0])
    directions = [(-1,-1), (-1,0), (-1,1),
                  (0,-1), (0,1),
                  (1,-1), (1,0), (1,1)]

    result = [[0] * n for _ in range(m)]
```

```
    for i in range(m):
        for j in range(n):
            live = 0

            for dx, dy in directions:
                x = i + dx
                y = j + dy

                if 0 <= x < m and 0 <= y < n:
                    live += board[x][y]

            if board[i][j] == 1:
                if live == 2 or live == 3:
                    result[i][j] = 1
            else:
                if live == 3:
                    result[i][j] = 1

    return result

import ast
board = ast.literal_eval(input())
print(gameOfLife(board))
```

```
= RESTART: C:/Users/miche/OneDrive/Desktop/DAA CLG LAB/11-10-26/Game of Life.py
[[1,1],[1,0]]
[[1, 1], [1, 1]]
>>>
```

3.

```
def champagneTower(poured, query_row, query_glass):
    dp = [[0.0] * (query_row + 2)
          for _ in range(query_row + 2)]

    dp[0][0] = poured

    for i in range(query_row):
        for j in range(i + 1):
            excess = max(0.0, dp[i][j] - 1.0)

            if excess > 0:
                dp[i + 1][j] += excess / 2
                dp[i + 1][j + 1] += excess / 2

    return min(1.0, dp[query_row][query_glass])
```

```
poured, query_row, query_glass = map(int, input().split())
print(champagneTower(poured, query_row, query_glass))
```

EXAMPLE 1:

```
= RESTART: C:/Users/miche/OneDrive/Desktop/DAA CLG LAB/11-10-26/Champagne Tower.
py
1 1 1
0.0
>>>
```

EXAMPLE 2:

```
= RESTART: C:/Users/miche/OneDrive/Desktop/DAA CLG LAB/11-10-26/Champagne Tower.
py
2 1 1
0.5
>>>
```